Warehouse Cold Chain & Inventory Management

Process Card Register

Purpose: Bridge the master Operational Process Map to the specific sales order.

Order Specification & Commercial Reference

Field	Order Detail	Field	Order Detail
Sales Order / PO	**LL628-0626**	**Scheduled Delivery**	06 June 2026
Commodity	Processed Edible Jellyfish	**Internal Barcode Serial**	9331100
Supplier Raw Lot	Jap682/1234	**Net Weight / Packaging**	15.00 kg 15 × 1 kg pouches
Import Inbound Date	01 May 2026	**Assigned Storage Slot**	Cold Storage Zone B R-04-12
Buyer / Client	Premier Supermarket Chain HK	**Inventory Rotation**	FEFO Expiry: 20 June 2026

Consolidated Process Card Register

Framework reference: Appendix C of the Operational Process Mapping Framework.

Register Field	Definition & Order-Specific Entry
Process Card Register ID	**Definition:** Unique tracking identifier for this storage and inventory lifecycle. **PC-WH-2026-9331100**
Master Step Reference	**Definition:** Corresponding macro-process step from the Main Process Map. **Warehouse Cold Storage & Inventory Management** (OPM-WH-CCIM01-v3)
Process Goals	**Definition:** Measurable parameters, storage criteria and SLA holding targets. • Maintain storage temperature between **+1.0°C and +3.5°C**. • Achieve **100% FEFO accuracy**, with Lot Jap682/1234 prioritized. • Maintain packaging vacuum and brine seal integrity: **zero loss**. • Picking handoff core temperature **≤ +4.0°C within ≤ 10 minutes**.

Risk Control Dimension	**Profile A: Standard Chilled / Frozen Whole Fish (**OPM-WH-CCIM01)	**Profile B: Critical / High-Risk RTE & Cured Invertebrates** (Order: LL628-0626)
Example Product:	Chilled or frozen whole fish	Edible jellyfish
Storage Temperature Band:	**Chilled:** 0.0°C to +4.0°C **Frozen:** ≤ −18.0°C	**Chilled saturated saline:** +1.0°C to +3.5°C — strict band
Thermal Action Limit:	Temperature > +5.0°C for > 30 minutes triggers review.	Temperature > +4.0°C for > 15 minutes, or > +6.0°C, triggers immediate QA lock.
Slotting & Access Rule:	Standard static racking; monthly cycle count.	Dedicated gravity-flow roller racks — physical FEFO lock.
Media / Brine Monitoring:	Visual ice-pack check.	Refractometer salinity check: ≥ 15° Baumé; zero vacuum loss.
WMS Release Interlock:	Soft barcode scan validation.	Hard digital interlock: system bars pick-wave if the lot is not the oldest.

Order-Specific Risks & Controls	**Definition:** Unique risks assessed for this high-risk, RTE salted-jellyfish lot. **Specific Risk 1 — RPS 16:** Extended storage duration: imported 01 May 2026 and picked 05 June 2026, representing 35 days in rack. Risk of desalination or slow Vibrio incubation if temperature drifts. **Specific Control 1:** Daily refractometer salinity test (≥ 15° Baumé) and multi-point infrared surface check.
Consolidated Level 1 In-Process Controls	**Definition:** Combined operational criteria, forms/rules and mandatory real-time execution records. **Criteria:** Profile B high-risk controls. • Storage envelope: **0°C to +4.0°C**. • Target: **+1.5°C**. • Maximum allowable excursion**: >4.0°C for >15 min**. • HACCP critical action limit: **> +6.0°C**. **Execution records:** • Real-time IoT temperature telemetry log: SYS-LOG-ZB-202606. • Daily rack inspection checklist: FORM-WH-RACK-03. • WMS slot scan log.
Consolidated Level 2 Outcome Controls	**Definition:** Combined final-release criteria, threshold limits and mandatory release documentation. **Criteria:** • Verified zero temperature excursions throughout the 35-day dwell. • Intact vacuum seals across all 15 pouches. • Sensory check confirms firm crunch and clean oceanic aroma. • Validated FEFO queue. **Release documentation:** • QA Pick-Release Endorsement Form — Stage 1 QA Sign-off on PO LL628-0626. • WMS Outbound Transfer Handshake Manifest.